BASE CHANGE COMPATIBILITY OF RHOM UNDER THE FOUR STANDARD HYPOTHESES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- The base change map for RHom is characterized by the adjunction formula relating it to the tensor-internal-hom diagonal map. invoked at proof line 28851. Matches the restricted map with the diagonal map of the next lemma.
- The tensor-internal-hom diagonal map K tensor RHom(M, L) -> RHom(M, K tensor L) is an isomorphism when K is perfect, R' is perfect over R, the map is flat with K pseudo-coherent and M bounded below, or R' has finite tor dimension with K pseudo-coherent and M bounded below. invoked at proof lines 28852-28853. Supplies the isomorphism in each of the four listed cases.

Source reference. Original equation 0E1X: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28801–28801. Referenced at statement line 28822 as the map whose isomorphy is claimed; the map is the subject inside the counted span.

Source reference. Original lemma 0BYN: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18036–18036. Cited at proof line 28847 to identify the restricted map with the diagonal map; the conclusion is used inside the counted span.

Source reference. Original lemma 0E1W: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28763–28763. Cited at proof line 28851 to match the left and right hand sides of the restricted map; the conclusion is used inside the counted span.

Source reference. Original lemma 0ATK: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28653–28653. Cited at proof line 28852 to conclude the isomorphism in each listed case; the conclusion is used inside the counted span.

Lemma 0.2. *Let $R \to R'$ be a ring map. Let $K, M \in D(R)$. The map (0E1X)*

$$R\operatorname{Hom}_R(K, M) \otimes^{\mathbf{L}}_R R' \longrightarrow R\operatorname{Hom}_{R'}(K \otimes^{\mathbf{L}}_R R', M \otimes^{\mathbf{L}}_R R')$$

is an isomorphism in $D(R')$ in the following cases

(1) *K is perfect,*
(2) *R' is perfect as an R-module,*
(3) *$R \to R'$ is flat, K is pseudo-coherent, and $M \in D^+(R)$, or*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(4) *R' has finite tor dimension as an R-module, K is pseudo-coherent, and $M \in D^+(R)$*

Proof. We may check the map is an isomorphism after applying the restriction functor $D(R') \to D(R)$. After applying this functor our map becomes the map

$$R\mathrm{Hom}_R(K, L) \otimes^{\mathbf{L}}_R R' \longrightarrow R\mathrm{Hom}_R(K, L \otimes^{\mathbf{L}}_R R')$$

of Lemma 0BYN. See discussion above the lemma to match the left and right hand sides; in particular, this uses Lemma 0E1W. Thus we conclude by Lemma 0ATK. □